Figure 8-1 The hierarchical approach examines all steps to complete the task.

While the primary reason to do learner task analysis is to identify the necessary steps to achieve the desired outcome, there is a valuable by-product: You don't waste training. Training time, seat time, and budgets are wasted when topics are included in the learning objectives that don't contribute to the tasks that learners will actually use in their roles and responsibilities. If you're teaching Microsoft Word and your class spends an hour on creating indexes, this hour is only valuable if the people in the class will actually use the technology to create indexes. You can do the math: If you have a dozen people in class, that's 13 hours wasted every time the instructor teaches indexes (the 12 participants and the instructor). Learner task analysis removes these nonvalue-added objectives and focuses on what needs to be taught for the benefit of the organization.

NOTE To actually do hierarchical analysis you define what the learner must know in order to do a given task. Each action necessary to achieve the end result is identified in a hierarchy.

Exploring Bloom's Taxonomy

Benjamin Bloom was an educational psychologist at the University of Chicago. His claim to fame is that he led a group of educators to develop a classification of learning goals for people. The actual taxonomy was developed by this group of educators but was named after Bloom, as he led the effort and edited the corresponding work *Taxonomy of Educational Goals*. The taxonomy has evolved since its release in 1956, with a significant update in 2000, but for the most part, stays true to its original concept: Define the objectives educators have for learners.

Traditionally Bloom's Taxonomy focused on pedagogical theories, though it's now a common theme in adult education. The taxonomy divides the learning objectives into three different domains: cognitive, affective, and psychomotor. Like much of adult education theory, Bloom's Taxonomy subscribes to the theory that to learn at a higher level of understanding you must first master the lower-level domains. This doesn't mean that educators should teach just to one domain at a time, but rather, they should embrace all three domains by teaching and structuring classes to allow participants to discover learning through experience and information transfer.

Originally, the *Taxonomy of Educational Goals* just focused on the cognitive domain. Over time, as mentioned, it evolved to include the affective and psychomotor domains. Within each domain there is a related taxonomy of terms that describe the actions within it. People are stronger in different domains based on their experience, maturity, education, and natural leanings. While in this book, and for your CompTIA CTT+ examination, you won't need to learn about the finer details of the taxonomy, you should be familiar with each domain and how it may affect you as a technical trainer.

VIDEO See the video *Learning Bloom's Taxonomy*.

Reviewing the Psychomotor Domain

The psychomotor domain describes the learning and application of physical tools and abilities—like installing a hard drive or punching down a network cable. Technically, the psychomotor domain describes the individual tasks, like using a screwdriver, opening the computer cover, removing the old hard drive, and so on. The psychomotor domain consists of the individual activities of the task that are mastered in order to achieve the end result.

The psychomotor domain describes the learners' skills, motor skills, manipulation of technology, ability to follow instructions, and eventually mastery of the physical activity through repetition. There are seven levels of performance within the psychomotor domain:

• **Perception** This attribute relies on the individual using the senses to determine what physical activity should happen. For example, a technician could smell wire burning and know how to respond while working on the computer; or a person could see a computer fan not spinning and test the power connectivity.

• **Set** Describes the readiness to respond to predictable situations. It's almost like an evolved mindset based on experience with a task. For example, a person could hear a noise in a printer that usually means a paper jam, and then the person could follow a checklist or directions to pull out the bottom tray, flip open the printer cover, reset the toner, and then shove the tray back in place to fix the problem.

• **Guided response** The learner can follow the instructor's directions to perform a motor skills task. For example, as an instructor you might coach a person through the process of installing a piece of hardware or completing a series of steps in a software program to reach a specific result. Once you've guided the learner through the process the learner will then become more efficient, and eventually master, the process through your continued guidance and repetition.

• **Mechanism** The guided responses are learned through repetition, and the participant understands each step of the guided responses and develops a mindset to completing the task. The mechanism is a slightly advanced combination of the set and the guided responses, where the learner can perceive the problem, operate with proficiency, and understand the correct responses to create the expected, predictable end result.

• **Complex overt response** With a bit more complexity than the previous mechanism the learner can complete a complex task based on their understanding and control of their motor skills. For example, a learner could use the mouse and keyboard in combination with a computer program to open, manipulate, edit the printer settings, and print a document. Several perceptions, sets, learned responses, and mechanisms are required to put all the individual tasks together into a hesitant-free execution of the desired result.

• **Adaptation** When a learner has mastered a specific skill, such as operating a printer, they can adapt their existing knowledge and apply it to new conditions and circumstances. For example, the learner could manipulate a different printer based on their existing knowledge of the current printer. A learner might also change the type of paper, the colors to be printed, the draft mode of the printer, and other settings based on their existing proficiency with a similar print device.

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• **Origination** This attribute happens when the learner takes their experience and expertise with existing knowledge and creates a new method, new skill, and expected outcome. For example, an individual could experiment in their photo-editing software to create new filters, new brushes, and effects and print to a new printer to produce a specific type of artwork.

In technology training you might make some assumptions that the people in your class have certain motor skills to complete the needed tasks. You might assume or observe that participants can handle a screwdriver, have the dexterity to open and close computer cases, and use the mouse and keyboard in combination. If you're teaching a beginning course, however, these assumptions may prove to be false. People who have never used the mouse before, installed any hardware, or created a network cable may find the motor skills difficult to master to get the expected results. Of course you'll be patient, coach the learners along, and allow them plenty of time to practice. Encourage your participants to keep trying, to keep experimenting when they're struggling with a complex task—with repetition they'll master the movements.

Reviewing the Affective Domain

Technology can be overwhelming to some people. I've taught classes, and perhaps you have too, where people have gotten teary in frustration with the technology. I've coached employees who have been so engrained with their current processes that the idea of moving from a paper-based workflow to an electronic workflow for data management has been so terrifying they've resigned. I've also worked with learners who have been so excited to embrace a new technology they rushed through exercises, worked through lunch hours, and didn't want to leave the classroom at the end of the day.

The affective domain describes, as in the preceding scenarios, the emotional values and feelings people attach to the technology they're learning. The affective domain is part of an individual's ability to learn, but it's also a significant part of the emotional health that allows a person to empathize with others, consider how a technical implementation may affect employees, and understand the frustration or excitement that others may attach to the technology. As a technical instructor you'll often be using the affective domain to recognize your learners' emotions and feelings toward the technology. The learners in your classes will often reveal their affective domains by expressing, sometimes with great emotion, how they love, hate, or despise the technology they're supposed to be learning.

NOTE I've taught classes, and I'm sure you have to, in organizations that are changing from one technology to a competing technology. In these instances you can clearly see the affective domain at work in the resistance to learning the new technology, as many people are emotionally attached to what they've already mastered.

Within the affective domain there are five stages that learners progress through as they react to the technical training. Figure 8-2 shows the five stages of learning in the affective domain, and you can follow along with this list:

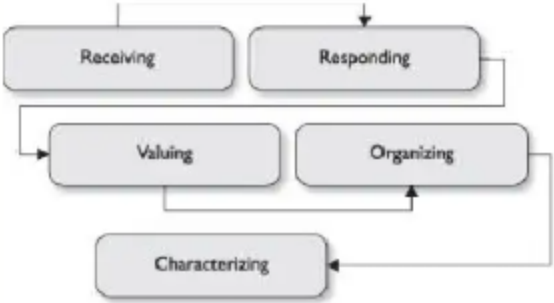

Figure 8-2 There are five stages to learning through the affective domain.

• **Receiving** This first stage of learning in the affective domain requires that the learner at least receive the information you're offering. The participant must, at a minimum, be passively involved in the learning process by being open to learning. If participants come to class with a predetermined decision to refuse to learn, then they've not even started the receiving process to accept the information.

• **Responding** Once a learner has passed through the receiving stage the individual will naturally enter the responding stage by *responding* to what's being presented to them in the learning environment. In responding, the individual participates in the learning process. This stage is evident when a person is asking questions, participating in discussions, completing exercises, and contributing to the course.

• **Valuing** Some components of your teaching will interest a learner more than other topics. When a person becomes really interested in what you're teaching, or they're excited or anxious about a topic, they're attaching value to the information you're teaching. For example, a person may really be excited to learn about configuring an e-mail server because it's what they'll do in their job. That same person, however, may be bored with the configuration of a printer server because they'll never likely do that work or they've already mastered that topic.

• **Organizing** As a person becomes more confident and experienced in a technology in your class, they'll begin to organize their thoughts, beliefs, and understanding to formulate their own ideas and expectations on outcomes. For example, a learner in a networking class may have an understanding of how a network operates, so they'll test their understanding by connecting and removing network cables to PCs and testing the outcomes to prove or disprove their beliefs and hypotheses about what may happen.

• **Characterizing** When an individual has learned information, they'll apply the information and change their behavior. The learner now has a value and an organized belief that directly affects how they behave and how they'll learn new and relevant information. For example, a person in a Microsoft Word class could have a goal to create formatting styles. They value this topic, as it directly affects their day-to-day work; they've organized their understanding based on what you've taught them, and now they'll continue to refine and experiment with formatting styles to become more proficient in applying this information to meet their particular needs as they fulfill their role.

The affective domain, in some theorists' opinion, is a filter to learning. If the individual has negative emotions about the subject matter they are to learn, then there may be a predetermined refusal to learn. Other theorists believe that people don't really dip into the affective domain until they've processed the information and made a determination about the effect of the learning topic, the technology in your case, and how the subject matter may alter a person's life. As a technical trainer, it's valuable to acknowledge the affective domain, as how people feel affect how they learn. If you can engage learners, which you will as part of a learner-centered environment, people will be more likely to accept the technology, be less intimidated by the material, and feel more comfortable learning something new.

Reviewing the Cognitive Domain

The final domain in Bloom's Taxonomy is the cognitive domain. The cognitive domain is often believed to be end result of the affective and psychomotor domains. As a technical trainer you'll often lecture on and demonstrate technology, with the goal of the learner absorbing the material and leaving the class with the necessary knowledge. Evidence of the cognitive domain is usually seen in the behavioral change a person exhibits based on their complete understanding and acceptance of the value of the education. For example, if you teach a class on how to create and administer a database, the people in the class will show evidence of the cognitive domain by applying their understanding in the workplace because they understand why the database should be administered and the plan to complete its administration.

The application of knowledge, however, isn't always a correct reflection of the cognitive domain. A person could still not understand how to operate with a given technology, but they can be intelligent enough to follow step-by-step instructions. The cognitive domain goes beyond the ability to just follow directions—it's the understanding and the application of the understanding to bring about a specific result. Through experience, practice, and repetition individuals show evidence of the depth of understanding and knowledge.

Here are the components of the cognitive domain, starting at the bottom of the hierarchy:

• **Knowledge** The fundamental element of the cognitive domain is knowledge, sometimes also referred to as remembering. This element is evident when a person can recall specific information, technical terms and concepts, and answer questions about the technology. For example, a person could explain how a computer connects to the Internet. Many facts, concepts, and ideas have to be expressed with this task analysis.

• **Comprehension** The second element is comprehension of the technical topic, sometimes referred to as understanding. Certainly there's a difference between just recalling facts and concepts and expressing an understanding of the technical information. This cognitive element is evident when a person can take a learned concept and then explain the concept in their own language. For example, a person would be able to explain the benefits of a wired network versus a wireless network.

• **Application** The application of knowledge is done by taking existing comprehension and applying it to new scenarios and problems. This is basically building on existing knowledge to discover new knowledge. For example, a person could complete a Network+ course and then troubleshoot a network problem based on what they've learned already. Imagine the person serving as a technician and going to a client's site where they've never worked before. By observing the problem, looking for

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causal factors, and then applying what they've learned, they can resolve the network connectivity issue.

• **Analysis** Analysis, in the cognitive domain, is demonstrated by examining a problem or scenario and thinking through the problem's root causes and causal factors. The participant can define what the problem is, describe the relationship between the causal factors and the problem to be solved, and break down the problem into an organized description. For example, a user may not be able to print to a network printer. The learner could break down the problem into three logical components: user's computer, the network, and the remote printer. Then, within each component, they analyze the conditions to determine where the actual problem, or related problems, may exist. Finally, the learner could draw a conclusion as to why the user can't print and how the problem can be resolved.

• **Synthesis** This component of the cognitive domain is sometimes referred to as create because the learner is creating a new solution based on what they've already learned. For example, by understanding how to program and work with Hypertext Markup Language (HTML) an individual can create many different websites. Obviously not all websites look the same, as it's the creativity, or the synthesis of knowledge, that allows people to create and customize webpages. In order to do this well, however, there must be knowledge, understanding, and application of information to put it all together and create a new webpage. Furthermore, several designers could be challenged to create a specific look and feel of a website, but each designer could take their own approach to create the same end result.

• **Evaluation** The final component in the cognitive domain is evaluation. This component is evident when a person can explain their logic and reasoning behind how they use the technology to solve a problem. Consider how two network engineers may have differing, and passionate, opinions on the best network operating system, the best hardware, and the best configurations for their clients. Both engineers may have reliable, usable results, but their beliefs and opinions have been shaped by their knowledge, understanding, and past application (experience). This is evident when they evaluate a problem or opportunity and present a solution that they believe is the preferred method of implementation.

Within the cognitive domain key action verbs are associated with each element. These verbs describe the type of action that demonstrates the element. As a technical trainer you can assess a participant's level of cognitive application by seeing evidence of these verbs in each element. If you're creating exercises and exams, you can choose a category and associated verb to test the learner's depth of comprehension in a given element. For example, a trainer may offer an exercise where the learner is to break down a network infrastructure and then categorize the types of network components. This exercise would test the learner's depth of cognitive application and synthesis. Table 8-1 is a summary of the cognitive elements and the associated verbs.

Knowledge	Comprehension	Application	Analysis	Synthesis	Evaluation
Arrange	Classify	Apply	Analyze	Arrange	Appraise
Define	Convert	Change	Appraise	Assemble	Argue
Describe	Defend	Choose	Break down	Categorize	Assess
Duplicate	Describe	Compute	Calculate	Collect	Attach
Identify	Discuss	Demonstrate	Categorize	Combine	Choose
Label	Distinguish	Discover	Compare	Comply	Compare
List	Estimate	Dramatize	Contrast	Compose	Conclude
Match	Explain	Employ	Criticize	Construct	Contrast
Memorize	Express	Illustrate	Diagram	Create	Defend
Name	Extend	Interpret	Differentiate	Design	Describe
Order	Generalized	Manipulate	Discriminate	Develop	Discriminate
Outline	Give example(s)	Modify	Distinguish	Devise	Estimate
Recall	Identify	Operate	Examine	Explain	Evaluate
Recognize	Indicate	Practice	Experiment	Formulate	Explain
Relate	Infer	Predict	Identify	Generate	Interpret
Repeat	Locate	Prepare	Illustrate	Plan	Judge
Reproduce	Paraphrase	Produce	Infer	Prepare	Justify
Select	Predict	Relate	Model	Rearrange	Predict
State	Recognize	Schedule	Outline	Reconstruct	Rate
	Review	Show	Point out	Relate	Relate
	Rewrite	Sketch	Question	Reorganize	Select
	Select	Solve	Relate	Revise	Summarize
	Summarize	Use	Select	Rewrite	Support
	Translate	Write	Separate	Set up	Value
			Subdivide	Summarize	
			Test	Synthesize	
				Tell	
				Write	

Table 8-1 The Cognitive Domain Includes Action Verbs for Each Cognitive Element

The cognitive domain is best described as the person's ability to understand and apply information. The depth of understanding and the accuracy of applying knowledge are often based on the experience and maturity of the person doing the application. As you might expect, a person who's new to technology and is taking a hardware configuration course may feel more overwhelmed with the terminology and concepts than a person who's worked in the technology arena for the past 20 years.

Chapter Summary

In adult education, and particularly in technical training, the focus is on the learners' ability to comprehend and apply the information you're offering. This concept of moving the focus from the trainer to the participants' needs is the heart of learner-centered instruction. Learner-centered instruction examines the needs of the participants, beginning with the course development, and is carried through to the instructor's delivery of the technical content. The intent is to help learners retain the technical knowledge and apply this information in their roles within the organization. This concept is also known as performance-based learning, because the people in your classes are learning how to change their behavior and perform at higher standards based on the correct information you've offered.

Adult learners must see the connection between what you're teaching and how it will affect their lives and job performance. If there's no inquiry into how the learners will apply the information in their lives and jobs, then there's a less realistic chance of creating learner-centered instruction. Commercial trainer centers and in-house training departments alike should examine what the learner needs to know in order to maximize performance in their roles. It's a waste of organizational

money and time to teach topics that won't be utilized or applied by the individual. This is why the instructor's wealth of observation, experience, and conversations with learners are all needed to fully capture and understand how the class can be structured, and sometimes modified, to best support the learners' post-class behavior.

In some cases the instructor must play the role of business analyst and instructional designer to fully capture the learners' needs. Analyzing learner tasks, ideally before class begins, gives the instructor insight into the processes, procedures, and expected outcomes of activities learners do in their roles. Shadowing the learners through active and passive observations helps the instructor experience and understand the work that learners will complete. Just understanding how a software program can work isn't enough for an instructor to understand how the learners will actually use the software. For example, an instructor can understand the functionality of Microsoft Excel, but learners can apply the functionality of Microsoft Excel for financial, calendar, and even data usage.

Associated with performance-based training is Bloom's *Taxonomy of Educational Goals*. This philosophy of education and knowledge retention helps define the objectives educators have for learners. Within the taxonomy there are three escalating domains of comprehension and mastery. First, the psychomotor domain defines the motor skills, eye-hand coordination, and manipulation of the technology, such as using a mouse and keyboard. Next is the affective domain that describes the emotional values learners attach to concepts, and sometimes technology, and how it affects their abilities to learn. The cognitive domain is the most robust of the three, as it describes the actual comprehension, application of information, and change in behavior based on what people have learned in your technical classes.

Key Terms

active observation Part of learner task analysis where the observer is actively involved in the task, asks questions for clarity, and may even participate in the actual work to better understand the tasks the learners will need to be able to complete with the technology.

adaptation A component of the psychomotor domain in Bloom's Taxonomy that happens when a learner has mastered a specific skill and can adapt their existing knowledge and apply it to new conditions and circumstances.

affective domain The emotional values and feelings people attach to learning; part of an individual's ability to learn, but it's also a significant part of the emotional health that allows a person to empathize with others.

analysis A component of the cognitive domain in Bloom's Taxonomy. Analysis in the cognitive domain is demonstrated by examining a problem or scenario and thinking through its root causes and causal factors.

application A component of the cognitive domain in Bloom's Taxonomy. Application of knowledge is expressed by taking existing comprehension and applying it to new scenarios and problems.

Benjamin Bloom Education psychologist at the University of Chicago; developed, along with a group of educators, the Taxonomy of Educational Goals to help define the objectives educators have for learners.

characterizing Part of the affective domain in Bloom's Taxonomy. When an individual has learned information they'll apply the information and change their behavior. The learner now has a value and an organized belief that directly affects how they behave and how they'll learn new and relevant information.

cognitive domain In Bloom's Taxonomy, the cognitive domain defines the actual knowledge gained and ability to act on understanding as a result of learning.

complex overt response Component of the psychomotor domain in Bloom's Taxonomy that happens when the learner can complete a complex task based on their understanding and control of their motor skills.

comprehension A component of the cognitive domain in Bloom's Taxonomy. Comprehension is sometimes referred to as understanding, and is expressed through an understanding of the technical information and the ability of the learner to explain this understanding.

evaluation A component of the cognitive domain in Bloom's Taxonomy. It is evident when a person can explain their logic and reasoning behind how they use the technology to solve a problem.

guided response Component of the psychomotor domain in Bloom's Taxonomy where the learner can follow the instructor's directions to perform a motor skills task.

hierarchical approach Breaks down tasks into a hierarchy of events, sometimes called subtasks, that contribute to the final result of the task or assignment.

knowledge A component of the cognitive domain in Bloom's Taxonomy. This element is evident when a person can recall specific information, technical terms, and concepts, and answer questions about the technology.

learner-centered environment The instructor puts the focus of the class on the learners and their need to change their behavior by applying the information gained.

mechanism Component of the psychomotor domain in Bloom's Taxonomy. The mechanism is a slightly advanced combination of the set and the guided responses, where the learner can perceive the problem, operate with proficiency, and understand the correct responses to create the expected, predictable end result.

organizing Part of the affective domain in Bloom's Taxonomy. As a person becomes more confident and experienced in a technology in your class, they'll begin to organize their thoughts, beliefs, and understanding to formulate their own ideas and expectations on outcomes.

origination Component of the psychomotor domain in Bloom's Taxonomy that happens when the learner takes their experience and expertise with existing knowledge and creates a new method, new skill, and expected outcome.

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